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Attention-guided bidirectional memory autoencoder for video anomaly detection

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ABSTRACT

Video anomaly detection is a critical task with applications in diverse domains, including security, transportation, and industry. However, building a robust model remains challenging due to the inherent ambiguity between normal and abnormal behaviors, and the subtle nature of abnormal events. To address these challenges, this paper introduces the Attention-guided Bidirectional Memory Autoencoder (ABMA). ABMA uses a novel Attention-guided Bidirectional Memory (ABM) module to store prototypical normal patterns in both forward and backward directions. Moreover, ABMA is trained in a supervised manner using both normal and pseudo-abnormal samples. This architecture allows ABMA to more effectively distinguish between normal and abnormal events, even in subtle cases. We evaluate ABMA on four widely used benchmarks: Ped2, Avenue, ShanghaiTech, and IITB-Corridor. Experimental results demonstrate that our approach outperforms all tested state-of-the-art methods in four scenarios including long-duration anomalies, short-duration anomalies, missing-blurred data and cross-dataset evaluation. Finally, the ablation study and visualization results further validate the advanced capabilities of ABMA.

1. Introduction

Video anomaly detection has broad applications in multiple domains, including public safety, traffic management, and healthcare monitoring [1–3]. For instance, it has been applied to detect traffic violations, identify abnormal activities in medical facilities, and recognize suspicious behaviors in public spaces. Despite its potential, video anomaly detection remains a challenging task due to several factors that limit the accuracy of models. One major difficulty lies in the ambiguity between normal and abnormal behavior, which arises from the inherent complexity of anomaly detection and its strong dependence on context. Another challenge is the subtlety of abnormal events, which may appear only briefly and are therefore easily overlooked. Furthermore, the scarcity of annotated abnormal data presents a fundamental challenge for model training and evaluation, thereby motivating the development of unsupervised approaches that learn normal patterns and identify deviations as anomalies.

Among these approaches, autoencoder-like architectures have shown particular promise, as they are effective at learning normal appearance and motion patterns while producing higher errors for abnormal ones. Two major training paradigms exist in this context. The first is reconstruction-based, which relies on the error between the input and reconstructed frames as an anomaly indicator [4–8]. The second is prediction-based, which uses the difference between the predicted and ground-truth future frames as the anomaly criterion [9–12]. Both paradigms commonly employ an encoder–decoder structure but differ in whether they reconstruct input frames or predict future frames. However, conventional autoencoders often generalize too well, reconstructing both normal and abnormal frames accurately and thus reducing anomaly discriminability [4].

To address this issue, some researches have attempted to integrate memory modules to store representative normal features [4,11,13–15] facilitating the difference between normal and abnormal. Specifically, the memory includes a set of vectors called the memory items. Each of the memory items represents typical or representative features that are learned from the normal samples during training. In the testing phase, the incoming frames are matched against the stored patterns in memory. Because the memory module is trained to represent typical normal patterns, normal frames closely match the stored patterns, allowing accurate reconstruction. In contrast, anomalies deviate, resulting in higher reconstruction errors. The memory module effectively prevents AEs from generalizing to anomalies and enhances feature

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separability. Additionally, it mitigates overfitting by enforcing structured constraints, thus preventing the model from inadvertently learning abnormal patterns.

Although several methods have integrated memory modules into autoencoders (AEs) to improve the distinction between normal and abnormal events, none have leveraged the combined strengths of a bidirectional memory module and an attention mechanism. Furthermore, the ability of such models to detect subtle anomalies has not been thoroughly investigated. To address this gap, we propose the Attention-guided Bidirectional Memory Autoencoder (ABMA), a novel model designed to enhance anomaly detection performance. The motivation for combining the bidirectional memory module and the attention mechanism is to harness the potential complementary strengths of both components in anomaly detection. Specifically, the bidirectional memory module enables the model to capture and learn rich and meaningful spatio-temporal features from both past and future directions. Meanwhile, the attention mechanism allows the model to focus more effectively on the most informative and discriminative features produced by the bidirectional memory, suppressing irrelevant background information. Therefore, the combination of these two modules enables ABMA not only to learn rich normal features but also to focus more on the important features. Together, these components enable ABMA to more effectively identify subtle anomalies that are often overlooked by previous approaches. Here, the term "subtle anomalies" refers to abnormal events that are difficult to distinguish from normal ones—for example, anomalies that are context-dependent, partially occluded, or occur only for a very short duration. In this paper, we particularly focus on investigating short-duration anomalies, as such events can be systematically generated from the original datasets. In addition, we adopt the skip-frame technique [8] to synthesize artificial anomalies, allowing the model to be trained in a supervised manner.

The novelties of ABMA can be summarized in three main aspects as follows:

- First, ABMA is the first model that proposes a bidirectional memory module to learn the rich representative of normal samples in both temporal directions.
- Second, ABMA combines the strength of the bidirectional memory module with a spatial attention mechanism that allows the model to focus more on important features that support the distinction between normal and abnormal events.
- Third, ABMA introduces a novel loss for bidirectional memory training with pseudo-anomalies and an anomaly scoring function to detect abnormalities in testing frames.

We conduct extensive experiments on four benchmark datasets and in four different scenarios to validate the effectiveness of the proposed model. The experimental results demonstrate that ABMA outperforms several state-of-the-art models. Further analysis through ablation studies and visualization helps us better understand the advantages of ABMA. The remainder of this paper is structured as follows. Section 2 reviews related works. Section 3 details the proposed methodology, including architectural components and key mechanisms. Section 4 outlines the experimental settings. Section 5 presents and analyzes the performance results. Section 6 conducts an ablation study to assess the impact of ABMA's components. Section 7 provides a visualization-based analysis to further interpret the model's behavior. Finally, Section 8 concludes the paper and suggests some future research directions.

2. Related work

This section provides a brief overview of previous research on video anomaly detection. In particular, it offers a detailed analysis of prominent studies on memory-augmented autoencoders and bidirectional prediction models. Subsequently, we compare the proposed ABMA model with several closely related approaches.

2.1. Memory-augmented autoencoders

Autoencoder-based approaches are widely employed for video anomaly detection due to their capability to reconstruct normal patterns. However, a key challenge is their tendency to accurately reconstruct both normal and abnormal patterns, which can lead to anomalies being misclassified as normal. To mitigate this limitation, memory modules have been integrated into autoencoder architectures to enhance the differentiation between normal and anomalous patterns [4,11,13–15]. Notable among these memory-augmented methods are MemAE [4] and MNAD [11], which serve as foundational models, each employing distinct memory mechanisms to model normality more effectively.

MemAE employs a memory module updated via backpropagation, which retains prototypical normal patterns during training and keeps them fixed during inference. By using a 3D convolutional network (Conv3d) to extract spatio-temporal features from video frames, this model effectively captures motion and structural information. Despite its effectiveness, MemAE requires a relatively large memory size and struggles with generalization in new environments [16]. MNAD [11] further refines these memory mechanisms by introducing feature compactness and separateness losses, thereby enforcing diverse and discriminative memory items. Unlike MemAE, MNAD adopts 2D convolutions, stacking multiple frames as channels, and dynamically updates its memory during inference.

Recently, Li et al. proposed ATT-FFP [17], an attention-based future frame prediction network that integrates a memory addressing module and a squeeze-and-excitation (SE) block [18] to improve feature selection. ATT-FFP predicts future frames conditioned on several consecutive inputs and leverages a combination of optical flow-based temporal losses, spatial reconstruction losses, and adversarial losses to enhance anomaly detection performance. Wang et al. introduced MAAM-Net [14], a memory-augmented appearance-motion network, where the memory module bridges appearance and motion feature representations. To strengthen the discrimination between normal and abnormal events, MAAM-Net employs a margin-based latent loss and introduces patch-based detection strategies. Cheng et al. presented NLR-VAD [19], a biologically inspired model that improves spatial-temporal normality learning via the Normality Learning Reinforcement Unit (NLRU), which uses diffusion models to refine irrelevant features and improve spatial detail perception. Furthermore, a cross-NLRU fusion module and a Memory-Augmented Autoencoder jointly capture fine-grained spatial–temporal correlations.

Some researchers have also incorporated multiple memory modules to better capture the patterns of normal frames. For instance, Cai et al. proposed AMMC-Net [13], which employs two memory banks to model appearance and motion features simultaneously. By maintaining consistency between these two modalities, AMMC-Net enhances the robustness of normal feature modeling. More recently, Zhao et al. combined local and global learning in video anomaly detection by integrating a spatio-temporal LSTM branch with a memory-prototype branch, enabling the model to capture both short-term dynamics and long-term normality prototypes [20].

2.2. Bidirectional prediction models

To better capture the temporal dependencies between frames in both directions, several bidirectional prediction models have been proposed that leverage both past and future frames to predict a target frame. For example, Chen et al. introduced a dual U-Net model that predicts the same target frame from both preceding and succeeding frames [21]. Fang et al. formulated video anomaly detection as a mutual supervision problem between two Siamese subnetworks, each modeling forward and backward temporal patterns [22]. Zhang et al. proposed a Mutual Learning Inspired Prediction Network (MLIPN) [23], which consists of two independent generators for unidirectional frame prediction and a shared discriminator.

Table 1
Comparison table of ABMA, MNAD [11], DEDDNet [24] and PB [8].

Criterion	ABMA	MNAD [11]	DEDDnet [24]	PB [8]
Objective	Learn more discriminative normal patterns to better detect anomalies, especially subtle or short-duration ones.	Learn representative normal patterns via memory to hinder anomaly reconstruction.	Learn bidirectional spatio-temporal features and detect anomalies via multi-scale evaluation.	Limit anomaly reconstruction by training with pseudo-anomalies.
Architecture	An Encoder, an Attention-guided Bidirectional Memory, and a Decoder.	An Encoder, a Memory Module, and a Decoder	Dual Encoder–Decoder with Feature Fusion Attention.	An 3D convolution Encoder-Decoder network.
Loss function	Normal loss (Intensity, Compactness, and Separateness loss) and Abnormal loss (negative Intensity and negative Compactness loss).	Reconstruction loss, Compactness loss, and Separateness loss.	Appearance loss and Motion loss.	Normal loss (Reconstruction loss) and Abnormal loss (negative Normal loss).
Anomaly score	It is the sum of the normalized image- and latent-space distances, each computed as the maximum of past and future scores.	It is the sum of normalized distances in the image and latent spaces in future direction.	It is computed based on a multi-scale anomaly evaluation strategy using prediction errors from both directions.	It is the difference between the reconstructed frame and the input frame, measured using PSNR.
Novelties	A bidirectional memory module with attention mechanism and a new loss function.	A novel compactness and separateness loss for memory-augmented AE	A bidirectional AE architecture and a multi-scale anomaly evaluation method.	An integration of pseudo-anomalies into the training process.

Recently, Zhong et al. developed a bidirectional spatio-temporal feature learning model, DEDDnet, which employs dual encoder-decoder structures and fuses forward and backward features using attention-based modules [24]. A multi-scale anomaly evaluation strategy is then applied to effectively detect abnormal events. Li et al. proposed a Multi-Branch GAN-based framework (MGAN-CL) [25], which utilizes dual generators to predict and reconstruct the same frame from past and future sequences. The authors also incorporated a pseudo-anomaly module to enrich the discriminator with outlier data. Lyu et al. introduced the Bidirectional Skip-frame Prediction (BiSP) network [26], a dual-stream autoencoder that skips frames during training to predict both forward and backward frames.

In the following sections, we propose a novel frame-level anomaly detection model named ABMA (Attention-guided Bidirectional Memory Autoencoder). Unlike previous works [21–26], which mainly rely on dual-stream models trained exclusively on normal data, our approach employs a single autoencoder architecture trained on both normal and pseudo-abnormal data. Moreover, ABMA differs from earlier memory-based models [13,14,17,27] by integrating a bidirectional memory augmented with attention to store and retrieve important latent representations that are critical for predicting future and past frames. This design enables ABMA to better exploit the temporal context and distinguish subtle anomalies through the joint modeling of spatial and temporal dependencies.

2.3. Comparative analysis of ABMA and related models

To better clarify the novelty of our proposed ABMA model, we compare it with three models, including MNAD [11], DEDDnet [24], and PB [8]. These models are typical models of the three main categories in video anomaly detection: memory-augmented models (MNAD), bidirectional prediction models (DEDDNet), and pseudo-anomaly-based models (PB).

It can be seen from Table 1 that ABMA introduces several key improvements over existing models. First, unlike MNAD, which focuses on learning representative normal patterns through a memory module, ABMA is designed to learn more distinguishable normal patterns using a bidirectional memory model. Architecturally, ABMA integrates a bidirectional memory with spatial attention between an encoder and a decoder, enabling more effective temporal and spatial feature learning compared to the single-direction memory used in MNAD. Second, although both ABMA and DEDDNet employ bidirectional learning schemes, they differ in how this is achieved: ABMA utilizes a novel bidirectional memory module augmented with an attention mechanism, whereas DEDDNet adopts a dual encoder–decoder architecture. Moreover, ABMA's loss function jointly considers both normal and abnormal components, providing a more comprehensive optimization objective than that of DEDDNet. Third, while ABMA also leverages pseudo-anomalies as in PB, its architecture using a dual memory module and spatial attention facilitates better discrimination between normal and abnormal events, leading to improved detection performance.

3. Methodology

The proposed model called Attention-guided Bidirectional Memory Autoencoder (ABMA) integrates an attention-guided bidirectional memory mechanism within an autoencoder. This architecture enables the simultaneous prediction of both past and future frames. Furthermore, we employ the skip-frame technique to synthesize pseudo-anomalies, which allows the model to be trained in a supervised manner. One potential issue with adding pseudo-anomalies to the training dataset is that the model may become biased toward these synthetic anomalies, thereby reducing its ability to detect real-world anomalies. To mitigate this limitation, we have evaluated the effectiveness of different types of pseudo-anomalies and identified the skip-frame type as the most beneficial, as it closely simulates real-world anomalies involving moving objects (see Section 6.2). Furthermore, since the features of pseudo-anomalies are not updated into the memory module (refer to Section 3.2 for details), the model's tendency to overfit on pseudo-anomalies is effectively reduced. As illustrated in Fig. 1, ABMA consists of three core components: an encoder, an Attention-guided Bidirectional Memory (ABM) module, and a decoder.

The most innovative component in ABMA is the Attention-guided Bidirectional Memory (ABM) module. The motivation for adopting a bidirectional structure stems from our observation that the transition and movement of normal objects are often smooth in both temporal directions, while abnormal events usually disturb this continuity. Therefore, by jointly reconstructing past and future frames, the model is encouraged to maintain temporal consistency under normal conditions and address inconsistencies associated with anomalies. Furthermore, the attention mechanism applied in ABM not only enhances the model's ability to highlight subtle deviations caused by anomalies but also improves the overall quality of frame reconstruction by guiding the decoder to attend to semantically meaningful regions. In the following subsections, we discuss the main components of ABMA in detail.

3.1. Encoder

The Encoder is designed to transform an input video sequence into a compact latent representation. The input of the Encoder includes both normal and pseudo-anomalous samples. The normal samples are sampled directly from the normal dataset, while the pseudo-abnormal samples are generated by applying the skip-frames technique [8] on the normal dataset. The selection of normal and abnormal inputs is

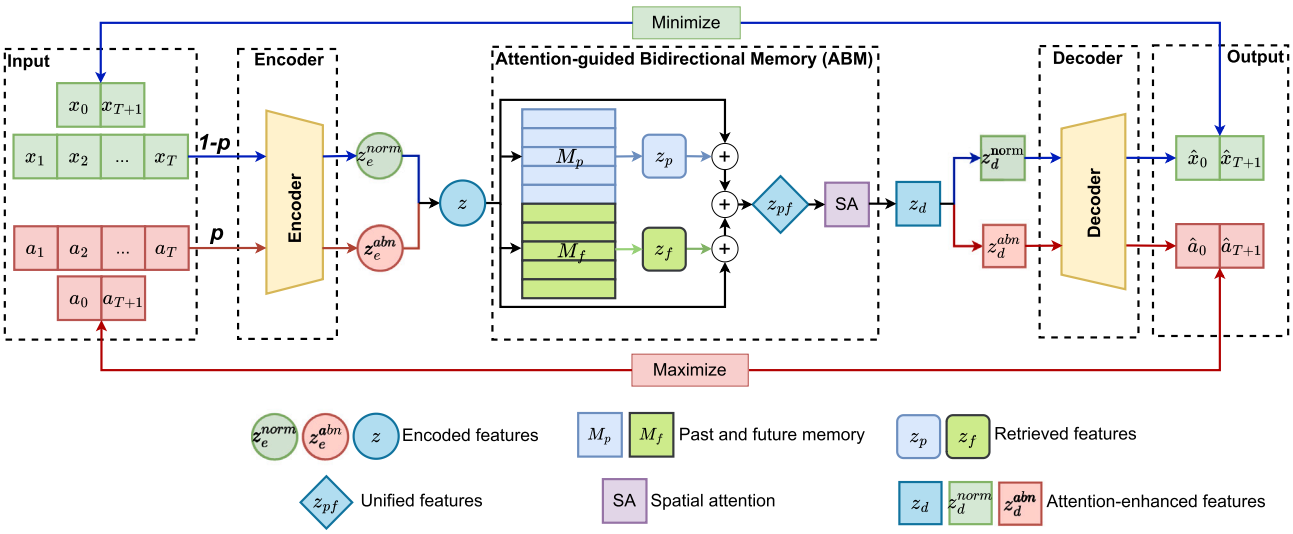

Fig. 1. Architecture of attention-guided bidirectional memory autoencoder (ABMA).

governed by a probabilistic mechanism p. This probabilistic sampling ensures a balanced mix of normal and pseudo-anomalous frames, thereby facilitating effective training of the ABMA.

Before encoding, each video frame is normalized to the range $[-1, 1]$, and the sequence is formed by concatenating the frames along the channel dimension to preserve temporal continuity. For a normal clip comprising $T + 2$ frames, denoted as $x = \{x_0, x_1, \dots, x_{T+1}\}$, the central subsequence $x_{1:T} = \{x_1, x_2, \dots, x_T\}$ is used as input to the model. The first and last frames, x_0 and x_{T+1}, represent the ground truth for the predictions of the past and future frames, respectively. The encoder f_e maps the normal sequence to a latent representation z_e^{norm} as follows:

$$z_e^{\text{norm}} = f_e(x_{1:T}) \tag{1}$$

Similarly, for a pseudo-anomalous sequence $a = \{a_0, a_1, \dots, a_{T+1}\}$, the same encoding strategy is employed. The encoder processes the central frames $a_{1:T} = \{a_1, a_2, \dots, a_T\}$, while a_0 and a_{T+1} serve as targets for the prediction of the past and future frames. The resulting latent representation z_e^{abn} is obtained as:

$$z_e^{\text{abn}} = f_e(a_{1:T}) \tag{2}$$

3.2. Attention-guided bidirectional memory (ABM)

The ABM module is a key component of the ABMA model, consisting of two main blocks: a bidirectional memory and a spatial attention mechanism. The bidirectional memory is designed to capture normal features from both past and future contexts, while the spatial attention mechanism enables the model to focus on the most relevant regions in the spatial domain. This architecture allows ABMA to more effectively distinguish between normal and abnormal events, particularly in subtle cases.

The **bidirectional memory** block is partitioned into the past subspace M_p and the future subspace M_f. Each memory subspace comprises N memory items of a fixed-length vector encoding prototypical patterns of normal behavior. The feature extracted by the encoder, z_e^{norm} or z_e^{abn}, is used to query the most relevant memory item in each memory. Since both z_e^{norm} and z_e^{abn} serve as inputs to the ABM, we collectively denote them as z_e for simplicity. The latent representation z_e is first partitioned into a set of Q vectors, denoted $\{z_j\}_{j=1}^{Q}$. Although both z_e^{norm} and z_e^{abn} interact with the memory subspaces in the same manner during the memory query process, only the normal features z_e^{norm} are employed in the memory update phase.

Update process: The memory items are periodically updated using the normal queries z_e^{norm} in a process similar to MNAD [11]. Let G_i denote the set of normal queries for which the memory item m_i is the nearest neighbor, then m_i is updated by performing a weighted aggregation of the associated query features, followed by L_2 normalization:

$$m_i = \left\| m_i + \sum_{j \in G_i} \tilde{v}_{ji} z_j \right\|_2 \tag{3}$$

where $\tilde{v}_{ji}$ is the normalized similarity weight between the query z_j and the memory item m_i:

$$\tilde{v}_{ji} = \frac{v_{ji}}{\max_{k \in G_i} v_{ki}} \tag{4}$$

where v_{ji} is the unnormalized similarity score obtained by computing the cosine similarity between the query and the corresponding memory item, followed by a softmax function applied over all queries:

$$v_{ji} = \frac{e^{(m_i)^T z_j}}{\sum_{k=1}^{Q} e^{(m_i)^T z_k}} \tag{5}$$

Query process: The items stored in the memory subspaces are used to form the new representations z_p and z_f corresponding to the past memory M_p and the future memory M_f by computing attention-weighted summations over memory items:

$$z_p = \sum_{i=1}^{N} w_{pji} m_{pi} \tag{6}$$

$$z_f = \sum_{i=1}^{N} w_{fji} m_{fi} \tag{7}$$

where w_{pji} and w_{fji} represent the weights between the query feature z_j and the memory items in M_p and M_f, respectively. These weights are calculated by applying the softmax function over the dot-product similarities as follows:

$$w_{pji} = \frac{e^{(m_{pi})^T z_j}}{\sum_{k=1}^{N} e^{(m_{pk})^T z_j}} \tag{8}$$

$$w_{fji} = \frac{e^{(m_{fi})^T z_j}}{\sum_{k=1}^{N} e^{(m_{fk})^T z_j}} \tag{9}$$

The representations z_p and z_f are concatenated with the encoded feature z_e resulting in memory-augmented features z_{pm} and z_{fm}. These

features are then concatenated to form a unified feature that captures the bidirectional temporal context.

$$z_{pf} = z_{pm} \oplus z_{fm} \tag{10}$$

The Spatial Attention (SA) module aims to extract informative regions from the unified feature z_{pf}. Given the unified feature $z_{pf} \in \mathbb{R}^{B\times C\times H\times W}$, where B is the batch size, C is the channel, and H and W are the height and width of the feature. This module computes two intermediate maps via channel-wise global average and max operations:

$$A_{\text{avg}} = \frac{1}{C}\sum_{c=1}^{C} z_{pfc} \tag{11}$$

$$A_{\max} = \max_{c=1,\dots,C} z_{pfc} \tag{12}$$

These maps are then concatenated and passed through a convolutional layer followed by a sigmoid function to obtain the spatial attention map $A \in \mathbb{R}^{B\times 1\times H\times W}$:

$$A = \sigma\left(\text{Conv2d}\left(A_{\text{avg}} \oplus A_{\max}\right)\right) \tag{13}$$

The final attention-enhanced output z_d is obtained via element-wise multiplication between the attention map and the input:

$$z_d = z_{pf} \odot A \tag{14}$$

This refined feature z_d, including the normal feature z_d^{norm} and the abnormal feature z_d^{abn}, is subsequently fed into the decoder, which uses it to simultaneously reconstruct the past frame and predict the future frame.

3.3. Decoder

The decoder f_d is responsible for generating past and future frames based on the attention-enhanced feature map, z_d. When the input sequence is a normal sequence, the decoder outputs two reconstructed frames, $\hat{x}_0$ and $\hat{x}_{T+1}$, corresponding to the past and future predictions, respectively.

$$\hat{x}_0, \hat{x}_{T+1} = f_d\left(z_d^{\text{norm}}\right) \tag{15}$$

Conversely, when the encoder receives a pseudo-anomalous input sequence, the decoder generates the corresponding predicted frames $\hat{a}_0$ and $\hat{a}_{T+1}$ as:

$$\hat{a}_0, \hat{a}_{T+1} = f_d\left(z_d^{\text{abn}}\right) \tag{16}$$

3.4. Objective functions

Two types of loss functions, namely normal loss and abnormal loss, are applied to train ABMA. These losses jointly encourage the model to preserve normality patterns while highlighting deviations that indicate anomalies. Fig. 2 presents the detailed losses of ABMA.

Normal loss: For a given normal sequence, the normal loss, $L_{\text{total}}^{\text{norm}}$, is designed to minimize the difference between the predicted and ground-truth frames. This loss combines two directional losses as follows:

$$L_{\text{total}}^{\text{norm}} = \sigma_p L_{p,\text{total}}^{\text{norm}} + \sigma_f L_{f,\text{total}}^{\text{norm}} \tag{17}$$

where σ_p and σ_f are two weighting coefficients. Each of the losses, i.e., the normal past loss, $L_{p,\text{total}}^{\text{norm}}$, and the normal future loss, $L_{f,\text{total}}^{\text{norm}}$, is the sum of three components.

$$L_{p,\text{total}}^{\text{norm}} = L_{p,\text{intensity}}^{\text{norm}} + \sigma_c L_{p,\text{compactness}}^{\text{norm}} + \sigma_s L_{p,\text{separation}}^{\text{norm}} \tag{18}$$

$$L_{f,\text{total}}^{\text{norm}} = L_{f,\text{intensity}}^{\text{norm}} + \sigma_c L_{f,\text{compactness}}^{\text{norm}} + \sigma_s L_{f,\text{separation}}^{\text{norm}} \tag{19}$$

The first component is the intensity loss $L_{p,\text{intensity}}^{\text{norm}}$ and $L_{f,\text{intensity}}^{\text{norm}}$ that measures the L_2 distance between the predicted and the ground truth

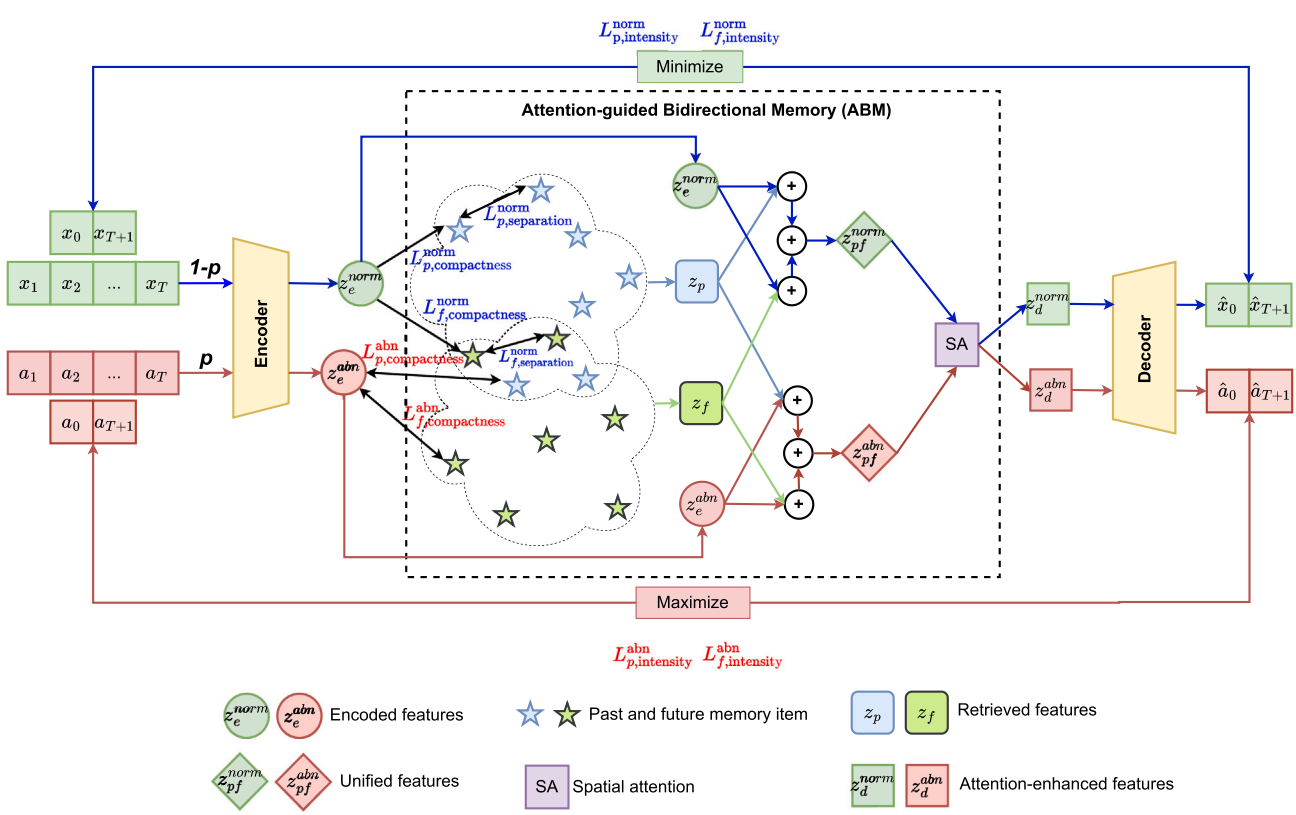

Fig. 2. Loss functions of ABMA. p and f denote 'past' and 'future', respectively, while 'norm' and 'abn' represent normal and abnormal properties.

frames.

$$L^{\text{norm}}_{p,\text{intensity}} = \left\|\hat{x}_0 - x_0\right\|_2 \tag{20}$$

$$L^{\text{norm}}_{f,\text{intensity}} = \left\|\hat{x}_{T+1} - x_{T+1}\right\|_2 \tag{21}$$

The second one is the compactness loss $L^{\text{norm}}_{p,\text{compactness}}$ and $L^{\text{norm}}_{f,\text{compactness}}$ that forces the memory item corresponding to the similar features to remain close, thereby enhancing the representation of normal patterns.

$$L^{\text{norm}}_{p,\text{compactness}} = \sum_{j=1}^{Q} \left\|z^{\text{norm}}_j - m_{pi}\right\|_2 \tag{22}$$

$$L^{\text{norm}}_{f,\text{compactness}} = \sum_{j=1}^{Q} \left\|z^{\text{norm}}_j - m_{fi}\right\|_2 \tag{23}$$

The third one is the separation loss $L^{\text{norm}}_{p,\text{separation}}$ and $L^{\text{norm}}_{f,\text{separation}}$ that prevents the memory items associated with the dissimilar features from collapsing together, maintaining a discriminative representation in the latent space.

$$L^{\text{norm}}_{p,\text{separation}} = \max\left\{0, \sum_{j=1}^{Q}\left(\left\|z^{\text{norm}}_j - m_{pi}\right\|_2 - \left\|z^{\text{norm}}_j - m_{po}\right\|_2 + r\right)\right\} \tag{24}$$

$$L^{\text{norm}}_{f,\text{separation}} = \max\left\{0, \sum_{j=1}^{Q}\left(\left\|z^{\text{norm}}_j - m_{fi}\right\|_2 - \left\|z^{\text{norm}}_j - m_{fo}\right\|_2 + r\right)\right\} \tag{25}$$

where m_{pi}, m_{fi} are the nearest items, m_{po} and m_{fo} are the second nearest ones of the query z^{norm}_j in the past memory and the future memory, respectively, r is a margin value, σ_c and σ_s are scale values.

Abnormal loss: The abnormal loss aims to maximize the discrepancy between the predicted frames and the ground-truth abnormal frames. The abnormal loss, $L^{\text{abn}}_{\text{total}}$, also consists of two components: the abnormal past loss $L^{\text{abn}}_{p,\text{total}}$ and the abnormal future loss $L^{\text{abn}}_{f,\text{total}}$ as follows:

$$L^{\text{abn}}_{\text{total}} = \sigma_p L^{\text{abn}}_{p,\text{total}} + \sigma_f L^{\text{abn}}_{f,\text{total}} \tag{26}$$

The losses $L^{\text{abn}}_{p,\text{total}}$ and $L^{\text{abn}}_{f,\text{total}}$ defined in Eqs. (27) and 28 include two terms, i.e., the negative intensity loss and the negative compactness loss, to prevent the model from reconstructing pseudo-anomaly inputs. The intensity loss and the compactness loss are defined in a similar way to the losses for normal input. The separation loss is excluded from the abnormal loss, as abnormal features are not updated in memory and thus do not contribute to the discriminative structure of the memory space.

$$L^{\text{abn}}_{p,\text{total}} = -L^{\text{abn}}_{p,\text{intensity}} - \sigma_c L^{\text{abn}}_{p,\text{compactness}} \tag{27}$$

$$L^{\text{abn}}_{f,\text{total}} = -L^{\text{abn}}_{f,\text{intensity}} - \sigma_c L^{\text{abn}}_{f,\text{compactness}} \tag{28}$$

3.5. Anomaly scores

The anomaly score is calculated for each frame based on the prediction error. For the i-th frame of a video, two metrics, i.e., the Peak Signal-to-Noise Ratio (P_i) and the feature distance (F_i), are calculated. The Signal-to-Noise Ratio is computed separately for the past and future directions, and the maximum value of the two directions is selected:

$$\text{P}_{pi} = 10\log_{10}\left(\frac{1}{\|x_{pi} - \hat{x}_{pi}\|_2^2/(H \times W)}\right) \tag{29}$$

$$\text{P}_{fi} = 10\log_{10}\left(\frac{1}{\|x_{fi} - \hat{x}_{fi}\|_2^2/(H \times W)}\right) \tag{30}$$

$$\text{P}_i = \max\left(\text{P}_{pi}, \text{P}_{fi}\right) \tag{31}$$

where $\hat{x}_{pi}$ and $\hat{x}_{fi}$ are the predicted frames in the past and future directions and x_{pi}, x_{fi} are their corresponding ground-truth frames. H is the height and W is the width of the frames.

The feature distance, F_i, is defined as the average L_2 distance between queries z_j of the frame i and their corresponding nearest memory items m_k. The maximum of the past and the future directions is calculated as the final value:

$$\text{F}_{pi} = \frac{1}{Q}\sum_{j=1}^{Q}\left\|z_j - m_{pk}\right\|_2 \tag{32}$$

$$\text{F}_{fi} = \frac{1}{Q}\sum_{j=1}^{Q}\left\|z_j - m_{fk}\right\|_2 \tag{33}$$

$$\text{F}_i = \max\left(\text{F}_{pi}, \text{F}_{fi}\right) \tag{34}$$

where Q is the number of queries. Both P_i and F_i are then normalized across all frames in the video using min-max normalization:

$$\mathcal{N}(v_i) = \frac{v_i - v_{\min}}{v_{\max} - v_{\min}} \tag{35}$$

where $v_{\min}$ and $v_{\max}$ denote the minimum and maximum values of the metric v over the entire video.

Finally, the anomaly score A_i of the i-th frame is a weighted sum of the normalized inverse P_i and the normalized F_i:

$$A_i = \alpha\left(1 - \mathcal{N}(\text{P}_i)\right) + (1 - \alpha)\mathcal{N}(\text{F}_i) \tag{36}$$

where $\alpha \in [0, 1]$ is a parameter that balances the contributions of image-level and feature-level errors.

4. Experimental settings

This section presents the structure of ABMA and the datasets used in the experiments.

4.1. Structure of ABMA

Inspired by U-Net [28] and MNAD [11], we employ two-dimensional convolutional (Conv2d) layers in both the encoder and the decoder to learn appearance features within individual frames and motion features across consecutive frames. The structure of ABMA is presented in detail in Table 2.

The encoder takes T consecutive RGB frames $x_{1:T}$ of size 3×256×256, concatenated into a T-channel tensor and normalized to $[-1, 1]$. The value of T is set to 4 for Ped2 to capture sufficient temporal context in its low-resolution videos. Conversely, $T = 3$ is used for Avenue, ShanghaiTech, and IITB-Corridor to balance model performance and computational efficiency in more complex scenes. The structure of the encoder is similar to the encoder in MNAD [11] that consists of four convolutional layers. At the final layer, the encoded feature z_e with shape (512 channels, 32×32) is output and passed to the next module, i.e., the ABM module.

The ABM module uses the vector z_e to query both the past and future memory by retrieving the nearest items based on similarity. The retrieved vectors from the past and future memory are concatenated to produce a memory-augmented representation of 2048 channels (32×32). After that, the average pooling and the max pooling are applied to produce a single-channel map of size (1, 32 ×). These two maps are then concatenated into a tensor of shape (2, 32×32), followed by a convolutional layer with a $k \times k$ kernel and a sigmoid function to produce the final attention map of size (1, 32×32). The kernel size is set at 3×3 for Ped2 [29], and 7×7 for Avenue [30], ShanghaiTech [31], and IITB-Corridor [32]. This attention map is subsequently element-wise multiplied with the original input, enabling the model to modulate each spatial location across all 2048 channels based on its importance.

The decoder reconstructs the past frame x_0 and the future frame x_{T+1}. The structure of the Decoder is also similar to the structure of the decoder in MNAD [11] except that it has two heads Conv2d(3,1,1)-Tanh to generate the past and future outputs with pixel values in $[-1, 1]$.

Table 2
The structure of the ABMA model includes an encoder, a decoder, and an attention-guided bidirectional memory (ABM) block.

Layer	Operators	Output
input	$x_1 \oplus x_2 \oplus \cdots \oplus x_T$	$x_{1:T}$: $B \times 3 *\mathrm{T} \times 256^2$
inc	$C \times 2$	$B \times 64 \times 256^2$
down1	MaxPool2d(2), $C \times 2$	$B \times 128 \times 128^2$
down2	MaxPool2d(2), $C \times 2$	$B \times 256 \times 64^2$
down3	MaxPool2d(2), C, Conv2d(3,1,1)	z_e: $B \times 512 \times 32^2$
Mp	Read(M_p, z_e), Update(M_p, z_e)	z_p: $B \times 512 \times 32^2$
Mf	Read(M_f, z_e), Update(M_f, z_e)	z_f: $B \times 512 \times 32^2$
zpf	$(z_e \oplus z_p) \oplus (z_e \oplus z_f)$	z_{pf}: $B \times 2048 \times 32^2$
avg	avg(z_{pf})	A_{avg}: $B \times 1 \times 32^2$
max	max(z_{pf})	A_{max}: $B \times 1 \times 32^2$
concat	$A_{avg} \oplus A_{max}$	A_{concat}: $B \times 2 \times 32^2$
att	Conv2d(7,1,3), Sigmoid	A: $B \times 1 \times 32^2$
Mout	$z_{pf} \odot A$	z_d: $B \times 2048 \times 32^2$
dec	$C \times 2$	$B \times 512 \times 32^2$
up1	$C\mathcal{T}$, $C \times 2$	$B \times 256 \times 64^2$
up2	$C\mathcal{T}$, $C \times 2$	$B \times 128 \times 128^2$
up3	$C\mathcal{T}$, $C \times 2$	$B \times 64 \times 256^2$
output1	Conv2d(3,1,1), Tanh	$\hat{x}_0$: $B \times 3 \times 256^2$
output2	Conv2d(3,1,1), Tanh	$\hat{x}_{T+1}$: $B \times 3 \times 256^2$

Notation:
– z_e is the encoded feature; z_d is the memory-enhanced feature fed into the decoder.
– Conv2d(3,1,1) is a 2D convolution with kernel = 3 × 3, stride = 1, and padding = 1.
– C = [Conv2d(3,1,1), BN, ReLU]; $C\mathcal{T}$ = [ConvTranspose2d(3,2,1), BN, ReLU].
– B denotes the batch size; BN stands for the batch normalization 2d (BatchNorm2d).
– $\oplus$ denotes the channel-wise concatenation.
– $\odot$ denotes the element-wise multiplication broadcast across all channels.

4.2. Datasets

To assess the effectiveness of the ABMA model, we conduct experiments on four video anomaly detection datasets, namely Ped2 [29], Avenue [30], ShanghaiTech [31], and IITB-Corridor [32]. The datasets differ significantly in scale, scene diversity, and the complexity of anomaly types, offering a comprehensive evaluation of the model's robustness under varying conditions.

Ped2 and Avenue are considered relatively simple datasets, both consisting of videos captured from a single static camera in a fixed environment. Ped2 features low-resolution footage and a small number of simple anomalies, such as bicycles entering pedestrian-only areas. Avenue, in contrast, presents a moderate increase in difficulty, with higher resolution and more diverse abnormal behaviors such as running and sudden movements, along with occlusions and slight camera motion.

ShanghaiTech (Sht) is a significantly more challenging benchmark, consisting of 330 training and 107 testing videos captured from 13 distinct campus scenes. It contains 274,515 training frames and 42,883 testing frames. The different environments, including outdoor areas, corridors, and open spaces, present significant challenges in learning generalizable representations. Anomalies mainly involve individual actions or object-related behaviors such as throwing, jumping, climbing, loitering, and riding bicycles.

The IITB-Corridor poses a more challenging task despite its simpler setting. It contains 208 training and 150 testing videos, with a larger volume of frames: 301,999 for training and 181,567 for testing. The videos, recorded in a single fixed indoor corridor, include 10 types of anomalies, encompassing individual, group, and static-object scenarios. Compared to ShanghaiTech, the IITB-Corridor presents a greater challenge due to the complexity of its anomalies, particularly group behaviors and static object scenarios, increasing the difficulty of behavior and semantic differentiation in anomaly detection.

4.3. Training phase

The ABMA model is implemented and trained end-to-end using the PyTorch library [33].[1] The experiments are carried out on a system equipped with an Nvidia RTX 4070 Ti graphics card with 16GB of memory.

Each original video in the normal training set is divided into clips of five consecutive frames. In each clip, the middle three frames are used as the model input, while the first and last frames serve as the labels. The frame window is then shifted forward by one frame until the end of the video. The clips from all normal videos are combined to form the normal training set. The clips are subsequently shuffled and organized into mini-batches, which are iteratively processed within each epoch. The clips for the pseudo-anomalous videos are generated in a similar manner. For every normal clip in a batch, the decision to replace it with a pseudo-anomalous sample is governed by an injection probability p. In our experiments, p is set to 0.001. If a uniformly generated random value is lower than p, a pseudo-anomaly is applied to the clip.

Optimization is performed using the Adam optimizer [34] with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The batch size is set to 4, and the number of training epochs is configured to 60 for Ped2 [29] and Avenue [30], and 10 for ShanghaiTech [31] and IITB-Corridor [32]. The initial learning rate is 2×10^{-4} and is decayed using cosine annealing [35].

The two weighting coefficients for compactness loss and separation loss in Eqs. (18) and (19) are set as $\sigma_c = 0.1$, $\sigma_s = 0.1$. The hyperparameter r in the separation loss terms (Eqs. 24 and 25) is set to 1.0. In normal loss (Eq. 17) and abnormal loss (Eq. 26), we set $\sigma_p = 0.5$ and $\sigma_f = 0.5$. The skip-frame lengths of [2, 3, 4, 5][2] are applied to all four datasets.

The parameter α in Eq. (36) is selected via grid search. The optimal values are set to 0.6, 0.4, 0.9, and 0.5 for Ped2, Avenue, ShanghaiTech, and IITB-Corridor, respectively.

4.4. Testing phase

In inference, frame-level anomaly scores are calculated for each video and compared with the corresponding ground-truth labels in the test dataset. Following the standard protocol in video anomaly detection, we adopt the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) as the evaluation metric. Thus, a fixed threshold is not required during the evaluation. A higher AUC score signifies better performance of the model in detecting anomalies.

5. Performance analysis

This section evaluates the performance of ABMA and compares it with several state-of-the-art methods across four scenarios: long-duration anomalies, short-duration anomalies, missing-blurred data, and cross-dataset evaluation. It also analyzes the computational efficiency and model complexity of ABMA.

5.1. Long-duration anomalies

We consider the original datasets as representing long-duration anomalies, as the anomalous events in these datasets typically last for more than 10 frames. In these experiments, the tested methods are grouped into four categories: future prediction models, bidirectional prediction models, memory-based models, and pseudo-anomaly-based models. Table 3 reports the results on three datasets: Ped2, Avenue, and ShanghaiTech (Sht), in which the results of the previous models are drawn from the respective published papers.[3]

[1] The source code of ABMA is available at: https://github.com/lenhoanh/ABMA.git.
[2] This means that 2, 3, 4, or 5 frames are removed from the original sequence to generate a pseudo-anomaly sequence.
[3] The IITB-Corridor dataset is excluded from this comparison due to the lack of publicly available results.

Table 3
AUC scores (%) of the tested models: boldface indicates the best result and underlining denotes the second-best result in each group.

Model	Year	Ped2	Avenue	Sht
Future prediction models				
FFP [9]	2018	95.40	85.10	72.80
Ada-Net [36]	2022	90.30	–	70.00
STD [7]	2022	96.70	87.10	73.70
AST-Net [37]	2023	97.40	86.70	73.60
PYTEF [38]	2023	93.80	87.12	73.60
VADMamba:FP [39]	2025	97.90	86.20	74.40
Bidirectional prediction models				
SIGnet [22]	2020	96.20	86.80	–
BiPred [21]	2020	96.60	87.80	–
MLIPN [23]	2022	96.00	85.90	73.10
DEDDnet-Parallel [24]	2022	97.40	88.60	–
DEDDnet-Fusion [24]	2022	98.10	89.00	74.50
MGAN-CL [25]	2023	96.50	87.10	73.60
Memory-based models				
MemAE [4]	2019	94.10	83.30	71.20
MNAD [11]	2020	97.00	88.50	70.50
AMMC-Net [13]	2021	96.60	86.60	73.70
ATT-FFP [17]	2022	95.40	86.00	71.40
STC-Net [40]	2022	96.70	87.80	73.10
STO [41]	2023	98.40	86.10	73.20
DnA [42]	2023	97.20	86.80	74.00
MAMC [43]	2024	96.70	87.60	71.50
PAME [44]	2025	95.59	–	72.29
MemAtr [45]	2025	95.20	83.80	73.40
Pseudo-anomaly-based models				
OG [5]	2020	98.10	–	–
LNTRA:Patch [6]	2021	94.77	84.91	72.46
FastAno:Patch [12]	2022	96.30	85.30	72.20
PB:Patch [8]	2023	95.33	85.36	72.77
PB:Noise [8]	2023	97.78	82.11	72.02
PB:SF [8]	2023	98.44	87.10	73.66
WeakAE:Noise [46]	2024	94.57	83.23	73.23
BAM-Fore [27]	2024	99.00	85.60	74.90
FastAno++ [47]	2025	98.10	87.80	75.20
ABMA (ours)	2025	**99.02**	**89.11**	**75.51**

ABMA consistently achieves the highest AUC scores across all three datasets. On Ped2, it attains an AUC of 99.02%, slightly surpassing the next-best method, BAM-Fore [27], which records 99.00%. On Avenue, ABMA reaches 89.11%, outperforming the second-best method, DEDDnet-Fusion [24], by 0.11%. On the more challenging ShanghaiTech dataset, ABMA achieves the highest AUC of 75.51%, exceeding the runner-up, FastAno++ [47], by 0.31%.

Compared to future prediction approaches such as VADMamba:FP [39], PYTEF [38], and STD [7], which are based on predicting future frames, ABMA achieves significantly better performance, particularly on ShanghaiTech where ABMA is better than VADMamba:FP and STD by more than 1.11%. This result highlights their lack of robustness in complex scenarios.

Among bidirectional prediction methods, DEDDnet-Fusion [24] is the best-performing model, achieving AUC scores of 98.10% on Ped2, 89.00% on Avenue, and 74.50% on ShanghaiTech. However, ABMA surpasses these results, with improvements of 0.92%, 0.11%, and 1.01% on Ped2, Avenue, and ShanghaiTech datasets, respectively.

For memory-based methods, STO [41], MNAD [11], and DnA [42] achieve the best results on Ped2, Avenue, and ShanghaiTech datasets, respectively. Nevertheless, their performance remains lower than that of ABMA, with margins of 0.62%, 0.61%, and 1.51%, respectively.

Finally, FastAno++ [47] is the best model on Avenue and ShanghaiTech in the pseudo-anomaly-based group. The AUC score of FastAno++ is 87.80% and 75.20% on Avenue and ShanghaiTech, respectively. However, these values are still lower than those of ABMA (89.11%, 75.51%) on both datasets.

Table 4
AUC scores (%) of ABMA and other models on the short-duration abnormal datasets (-S). Boldface indicates the best result.

Model	Ped2-S	Avenue-S	ShanghaiTech-S	IITB-Corridor-S
MNAD [11]	80.70	85.28	67.42	61.07
AST-Net [37]	86.19	85.35	72.78	60.13
PB:SF [8]	92.79	83.87	75.40	59.98
ABMA (ours)	**93.50**	**87.98**	**76.31**	**68.95**

Overall, the ABMA model achieves the best performance among all evaluated methods. Its strong results across all datasets are primarily attributed to the novel attention-guided bidirectional memory autoencoder (see Section 6 for details). This design allows ABMA to more effectively capture temporal irregularities while preserving normal patterns during training, resulting in more robust anomaly detection.

5.2. Short-duration anomalies

ABMA is designed to be more robust in detecting subtle anomalies due to its architecture based on the bidirectional memory and spatial attention modules. To validate this hypothesis, we conduct additional experiments on short-duration anomaly datasets. Specifically, we construct new datasets, namely Ped2-S, Avenue-S, ShanghaiTech-S, and IITB-Corridor-S, derived from their original counterparts (Ped2, Avenue, ShanghaiTech, and IITB-Corridor). The suffix "-S" denotes versions in which the abnormal segments in the selected test videos have been shortened to include fewer anomalous frames, while the normal segments remain unchanged. The generation of the shortened test set is conducted in three main stages. First, a proportion of videos from the original test set, denoted as p_v, is randomly selected using a fixed random seed. Second, within each selected video, all normal frame sequences are preserved, while only abnormal segments longer than a predefined threshold n_a are modified. For each of these abnormal segments, a proportion p_r of abnormal frames is randomly removed to simulate shorter and more discontinuous abnormal events. [4] Finally, the processed videos and their revised labels are saved to construct a new shortened test set, which enables a systematic evaluation of model performance under short duration anomalies. In our experiments, we set $p_v = 1.0$, $n_a = 10$, and $p_r = 0.5$.

Table 4 compares the best AUC scores of ABMA with three representative models, MNAD [11], AST-Net [37], and PB:SF [8], on the short-duration anomaly datasets. These models are selected for two main reasons. First, they exemplify different design strategies that incorporate key components such as memory modules, attention mechanisms, or pseudo-anomaly generators. Second, their official implementations are publicly available, enabling a fair and reproducible comparison. It can be seen from this table that ABMA consistently achieves the highest performance across all datasets, demonstrating superior robustness under shortened abnormal conditions. In particular, ABMA improves the AUC by 0.71% on Ped2-S, 2.63% on Avenue-S, 0.91% on ShanghaiTech-S, and a remarkable 7.88% on IITB-Corridor-S compared to the strongest baseline. These results confirm the effectiveness of ABMA in handling subtle and temporally sparse anomalies, where other models experience a significant performance degradation.

5.3. Missing-blurred data

To evaluate the effectiveness of ABMA in real-world imperfect scenarios such as frame loss and motion blur, we created a Missing-Blurred

[4] Another method for removing frames to generate pseudo-abnormal videos is systematic frame deletion. However, this approach often produces uniform anomalous segments and may bias the model toward learning regular, repetitive anomalous patterns. In contrast, the random method more accurately simulates short-term and discontinuous real-world anomalies while maintaining data diversity, and is therefore selected in our experiments.

Table 5
AUC scores (%) of ABMA and other models on the missing-blurred (MB) datasets. Boldface indicates the best result.

Model	Ped2-MB	Avenue-MB	ShanghaiTech-MB	IITB-Corridor-MB
MNAD [11]	69.37	72.42	59.54	58.70
AST-Net [37]	73.68	74.04	63.76	60.67
PB:SF [8]	63.32	64.33	52.76	49.65
ABMA (ours)	**77.39**	**77.41**	**65.38**	**62.31**

Table 6
AUC scores (%) under cross-dataset evaluation. Boldface indicates the best result.

Pre-trained on ShanghaiTech		
Models	Avenue	IITB-Corridor
MNAD [11]	81.91	68.39
AST-Net [37]	81.07	68.28
PB:SF [8]	77.74	66.55
ABMA	**83.89**	**69.24**

(MB) test set for each original dataset. The process includes three steps: (1) randomly deleting frames with a probability of $p_d = 0.02$ to mimic sporadic frame loss; (2) removing one short consecutive sequence of up to three frames per video to create temporal gaps; and (3) applying motion blur to 10% of frames using a kernel size randomly chosen from $\{3, 5\}$ and an angle from $-15°$ to $15°$. The modified frames are saved separately, and ground-truth labels are adjusted accordingly. To ensure reproducibility, a random seed is fixed for consistent frame selection across runs.

We compare the proposed ABMA model with three representative approaches including MNAD [11], AST-Net [37], and PB:SF [8] on the MB datasets. As shown in Table 5, ABMA consistently outperforms existing methods across all four benchmarks, demonstrating strong robustness under degraded visual conditions. Specifically, ABMA achieves the highest AUC scores of 77.39%, 77.41%, 65.38%, and 62.31% on the Ped2-MB, Avenue-MB, ShanghaiTech-MB, and IITB-Corridor-MB datasets, respectively. Compared with MNAD and AST-Net, ABMA exhibits clear performance gains, highlighting the advantages of incorporating bidirectional memory and spatial attention in its architecture. PB:SF, which relies on pseudo-anomalies, performs significantly worse across all datasets, indicating its limited generalization to distorted inputs. Overall, these results confirm that ABMA maintains superior anomaly detection performance even when video quality is degraded by frame loss or motion artifacts.

5.4. Cross-dataset evaluation

To evaluate the generalization capability of ABMA in unseen environments, we conducted a cross-dataset experiment in which the model was trained on the ShanghaiTech dataset and directly tested—without any fine-tuning—on two distinct target datasets: Avenue and IITB-Corridor. The ShanghaiTech dataset is chosen because it includes diverse camera viewpoints and complex motion patterns, allowing models to learn robust representations of normal behaviors. For comparison, we also evaluated three representative models—MNAD [11], AST-Net [37], and PB:SF [8]—using their pre-trained versions under identical experimental settings. As shown in Table 6, ABMA consistently achieves the highest AUC scores across both target datasets. Specifically, ABMA attains an AUC of 83.89% on Avenue, outperforming MNAD, AST-Net, and PB:SF by 1.98%, 2.82%, and 6.15%, respectively. Similarly, on IITB-Corridor, ABMA achieves an AUC of 69.24%, surpassing MNAD, AST-Net, and PB:SF by 0.85%, 0.96%, and 2.69%, respectively. These results demonstrate that ABMA possesses stronger cross-domain generalization and robustness than the compared models.

5.5. Computational efficiency and model complexity

ABMA contains a total of 20.37 million parameters and requires 62.33 GFLOPs per forward pass. The inference speed reaches 93 FPS on an RTX 3070, 237 FPS on an RTX 4070Ti, and 394 FPS on an RTX 5090, demonstrating that the model is capable of near real-time processing. Regarding memory efficiency, the inference memory consumption is 320.19 MB with a batch size of 1, while the training memory consumption is 2042.02 MB with a batch size of 4.

Table 7 compares the AUC scores (%) and computational efficiency of ABMA with MNAD [11], PB:SF [8], and AST-Net [37] under identical conditions on an RTX 4070Ti GPU. Compared to MNAD, a single-memory unidirectional model, ABMA has about 30% more parameters (20.37M vs. 15.65M) and an 8% higher computational cost (62.33 vs. 57.49 GFLOPs), mainly due to added attention and dual-memory modules. As a trade-off, the inference speed decreases moderately from 281 FPS in MNAD to 237 FPS in ABMA, and memory usage rises slightly from 312 MB to 320 MB. Despite this, ABMA delivers notable AUC improvements of 2.02%, 0.61%, and 5.01% on Ped2, Avenue, and ShanghaiTech, respectively. These results confirm that the bidirectional memory and spatial attention mechanisms substantially enhance anomaly detection accuracy while maintaining real-time efficiency suitable for surveillance applications.

Compared to PB:SF, a lightweight 3D convolutional autoencoder, ABMA achieves a more balanced trade-off between accuracy and efficiency. Although PB:SF has only 5.98M parameters, its dense 3D convolutional structure results in substantially higher computational cost (142.25 GFLOPs), more than twice that of ABMA (62.33 GFLOPs). Consequently, PB:SF exhibits slower inference (170 FPS vs. 237 FPS) and greater training memory consumption (2724 MB vs. 2042 MB). In terms of accuracy, ABMA consistently surpasses PB:SF, achieving AUC improvements of 0.58%, 2.01%, and 1.85% on Ped2, Avenue, and ShanghaiTech, respectively. Compared with AST-Net, a much heavier model with over 150M parameters and 680 GFLOPs, ABMA delivers higher detection accuracy while using only one-seventh the parameters and one-tenth the computational cost. Overall, ABMA provides an excellent balance between accuracy, complexity, and runtime efficiency, making it a practical and scalable solution for real-time video anomaly detection.

6. Ablation studies

This section first analyzes the impact of the spatial attention (A), memory module (M), and pseudo-anomalies (P) on ABMA's performance. We then investigate the impact of various parameters including pseudo-anomaly types, the injection probability, the scale parameters, the past-future weights, the number of memory items and the anomaly score parameter to the effectiveness of ABMA.

Table 7
Comparison of AUC scores (%) and computational efficiency among MNAD [11], PB:SF [8], AST-Net [37] and ABMA. Boldface indicates the best result in each group.

Model	Ped2	Avenue	Sht	Params(M)	GFLOPs	Infer(MB)	Train(MB)	FPS
MNAD [11]	97.00	88.50	70.50	15.65	**57.49**	312.06	**1836.09**	**281**
PB:SF [8]	98.44	87.10	73.66	**5.98**	142.25	**272.25**	2724.27	170
AST-Net [37]	97.40	86.70	73.60	150.97	680.02	2122.98	12,359.17	26
ABMA	**99.02**	**89.11**	**75.51**	20.37	62.33	320.19	2042.02	237

Table 8
Average and maximum AUC score (%) over 10 runs of ABMA variants. Boldface indicates the best result, and underlining denotes the second-best result.

Model	Ped2 avg/max	Avenue avg/max	ShanghaiTech avg/max	IITB-Corridor avg/max
ABMA-AMP	94.80/95.33	85.10/86.40	70.95/71.73	65.56/67.69
ABMA-P	94.91/95.40	86.08/87.37	71.64/72.09	67.64/68.20
ABMA-M	97.00/98.39	86.44/87.51	72.13/72.81	67.85/68.34
ABMA-A	97.19/<u>98.48</u>	87.53/<u>88.59</u>	72.72/<u>73.45</u>	68.15/<u>68.57</u>
ABMA	98.00/**99.02**	88.16/**89.11**	73.52/**75.51**	69.34/**70.90**

Table 9
Average and maximum AUC scores (%) over 5 runs of ABMA with different pseudo-anomalies. Boldface implies the best result, and underlining indicates the second-best result in each group.

Model	Ped2 avg/max	Avenue avg/max	ShanghaiTech avg/max	IITB-Corridor avg/max
baseline	94.43/95.40	85.54/86.15	71.81/72.09	68.13/68.20
Single pseudo-anomaly				
ABMA:Noise	94.76/95.75	83.68/84.03	70.38/72.77	68.19/68.59
ABMA:RF	93.30/94.75	85.00/85.83	71.91/72.66	68.27/68.65
ABMA:KF	96.68/<u>98.68</u>	85.66/86.73	72.69/<u>73.06</u>	67.88/68.40
ABMA:Occlusion	94.83/95.94	86.42/<u>87.82</u>	70.83/72.52	68.12/68.24
ABMA:Illumination	94.12/94.80	85.73/87.46	72.29/72.81	68.75/<u>69.36</u>
ABMA:Patch	94.30/95.08	85.56/87.17	72.61/73.03	68.16/68.51
Combined pseudo-anomaly				
ABMA:SF→Patch	94.03/95.02	83.77/83.91	69.56/71.27	67.31/67.71
ABMA:SF⊕Patch[0.2:0.8]	94.89/95.97	85.90/86.90	72.37/73.11	68.06/68.42
ABMA:SF⊕Patch[0.5:0.5]	95.13/96.21	86.55/87.39	72.61/73.28	68.32/69.00
ABMA:SF⊕Patch[0.8:0.2]	95.30/<u>96.43</u>	87.36/<u>88.01</u>	72.94/<u>73.37</u>	68.42/<u>69.50</u>
ABMA:SF	98.24/**99.02**	88.16/**89.11**	73.78/**75.51**	69.71/**70.90**

6.1. Variants of ABMA

Table 8 presents an ablation study on the contributions of three key components in ABMA over ten runs: spatial attention (A), the dual memory module (M), and pseudo-anomalies (P). Specifically, five ABMA variants are compared: ABMA, the full model; ABMA-A, the model without spatial attention; ABMA-M, the model with only a single memory module; ABMA-P, the model trained solely on normal data without pseudo-anomalies; and ABMA-AMP, the model without all three components.

Overall, ABMA consistently delivers the best results across all datasets, confirming that the joint integration of A, M, and P is vital for robust video anomaly detection. Removing spatial attention (ABMA-A) degrades performance-for instance, its AUC drops to 87.53% on Avenue and 68.15% on IITB-Corridor, compared to 88.16% and 69.34% for ABMA. Similarly, using a single memory (ABMA-M) or excluding pseudo-anomalies (ABMA-P) reduces accuracy. ABMA-M achieves 97.00% on Ped2 and 72.13% on ShanghaiTech, while ABMA-P yields 86.08% on Avenue and 67.64% on IITB-Corridor, all lower than the full model.

The largest performance drop occurs in ABMA-AMP, where spatial attention, memory, and pseudo-anomalies are all removed. In this case, average AUCs fall from 98.00%, 88.16%, 73.52%, and 69.34% to 94.80%, 85.10%, 70.95%, and 65.56% on Ped2, Avenue, ShanghaiTech, and IITB-Corridor, respectively. In summary, the results confirm that each ABMA component contributes uniquely to anomaly detection. Removing any component causes a clear performance drop, while eliminating all three yields the most significant decline.

6.2. Types of pseudo-anomalies

This subsection investigates the impact of different techniques for generating pseudo-anomalies including noise addition (:Noise), repeat-frames (:RF), key-frames (:KF), skip-frames (:SF), object occlusion (:Occlusion), sudden illumination changes (:Illumination), and patch injection (:Patch). The noise-based method adds Gaussian noise ($\mu = 0$, $\sigma = 0.05$). Repeat-frames simulate slow motion by duplicating frames $r \in [2, 3]$. The key-frame method (KF-NP [48]) selects non-consecutive frame pairs as anomaly candidates. Occlusion masks 5%–30% of the frame area with black pixels, while illumination injection adjusts brightness by a random factor (0.5–1.5), both within the normalized range $(-1, 1)$. Patch injection applies SmoothMixS [49] to embed CIFAR-100 [50] objects into normal frames, with patch sizes in $[10, \beta_1 W] \times [10, \beta_1 H]$ ($\beta_1 = 0.5$) and random shifts of up to $\beta_2 = 25$ pixels. Each pseudo-anomaly type (Noise, RF, KF, Occlusion, Illumination, Patch) is injected with probability $p = 0.001$.

Two experiments are also conducted to combine SF with Patch, namely the explicit sequential combination (ABMA:SF→Patch) and the independent mixture (ABMA:SF⊕Patch). In ABMA:SF→Patch, SF is applied first and then Patch is added to the remaining clip, with a probability of $p = 0.001$. In ABMA:SF⊕Patch, SF and Patch are injected independently with a total probability of 0.001. Three variants are included: ABMA:SF⊕Patch[0.8:0.2], [0.5:0.5], and [0.2:0.8], reflecting different mixing ratios between SF and Patch.

The results are summarized in Table 9, where the baseline (ABMA-P) excludes pseudo-anomalies. Models incorporating pseudo-anomalies that simulate accelerated object motion, namely ABMA:KF, ABMA:SF, ABMA:Patch and ABMA:SF⊕Patch, consistently outperform the baseline across all datasets, with ABMA:SF achieving the highest AUC scores. In contrast, ABMA:Noise, ABMA:RF and ABMA:SF→Patch provide limited or even negative gains; for instance, the average AUC of ABMA:RF falls below the baseline by 1.13% on Ped2 and 0.55% on Avenue.

For ABMA:Occlusion, a modest gain is observed on Avenue (+1.67%), ranking second after ABMA:SF, while performance on other datasets remains nearly unchanged. ABMA:Illumination achieves the second-best AUC on IITB-Corridor (+1.16%) but shows only minor gains on Avenue and ShanghaiTech and a slight drop on Ped2. These results suggest that occlusion and illumination disturbances often resemble normal scene variations, reducing their discriminative effectiveness.

For the combination between SF and Patch, both variants, i.e., SF→Patch and SF⊕Patch, remain inferior to ABMA:SF. ABMA:SF→Patch leads to the most severe degradation, dropping from 88.16% to 83.77% on Avenue and from 73.78% to 69.56% on ShanghaiTech. ABMA:SF⊕Patch provides only slight improvements over ABMA:Patch (e.g., on Avenue, 87.36% for ABMA:SF⊕Patch[0.8:0.2] vs. 85.56% for ABMA:Patch), yet they remain inferior to ABMA:SF. These counterintuitive results can be explained by two main reasons. First, SF directly disrupts temporal continuity, which aligns well with ABMA's bidirectional temporal prediction objective and thus provides strong supervision, whereas Patch introduces only local spatial noise that contributes weakly to temporal modeling and can even interfere with SF, especially in the sequential setting. Second, combining heterogeneous pseudo-anomalies forces the model to suppress both temporal and spatial distortions simultaneously, leading to over-regularization of the memory module and weakening the separation between normal and abnormal patterns, an effect most pronounced in SF→Patch.

Table 10
Average and maximum AUC scores (%) over 5 runs of ABMA with different pseudo-anomaly injection probabilities p on the Ped2 and ShanghaiTech datasets. Boldface indicates the best result.

p	Ped2 avg/max	ShanghaiTech avg/max
baseline	94.43/95.40	71.81/72.09
0.0005	95.09/97.35	72.09/72.59
0.001	**98.24/99.02**	**73.78/75.51**
0.005	95.59/98.06	72.73/73.28
0.01	93.65/97.71	72.58/72.88
0.05	91.36/97.78	71.92/72.82
0.1	86.25/95.82	70.45/71.03
0.5	77.08/87.11	67.94/70.01

Table 11
Average and maximum AUC scores (%) over 5 runs of ABMA with different values of the scale parameters (σ_c, σ_s) on the Ped2 and ShanghaiTech datasets. Boldface indicates the best result.

(σ_c, σ_s)	Ped2 avg/max	ShanghaiTech avg/max
(1.0, 1.0)	95.15/95.53	72.63/72.91
(0.1, 0.1)	**98.24/99.02**	**73.78/75.51**
(0.01, 0.01)	96.48/98.03	73.12/73.77
(0.001, 0.001)	95.99/97.20	72.64/72.88

Overall, these results confirm that ABMA attains its highest performance under temporally pseudo-anomaly supervision (SF only), whereas combining heterogeneous perturbations may introduce component interference and over-regularization, thereby potentially undermining anomaly discrimination.

6.3. Pseudo-anomaly probabilities

Table 10 presents the sensitivity analysis of the pseudo-anomaly injection probability p on the Ped2 and ShanghaiTech datasets. The baseline corresponds to the model trained without pseudo-anomalies. The results show that ABMA achieves the best overall performance when $p = 0.001$, yielding the highest AUC scores of 99.02% on Ped2 and 75.51% on ShanghaiTech. A smaller probability such as $p = 0.0005$ also improves the results compared to the baseline, but the gains are less substantial. If p increases beyond this value, the performance gradually declines on both datasets, suggesting that an excessive number of pseudo-anomalies may distort the distribution of normal patterns and hinder effective representation learning. Overall, these findings indicate that a low and adequate injection probability, $p = 0.001$, provides the best performance for ABMA.

6.4. Scale parameters

Table 11 presents the performance analysis of ABMA under different values of the scale parameters (σ_c, σ_s), which control the relative influence of the compactness and separateness losses in both past and future memories. The best results are obtained when both parameters are set to $(0.1, 0.1)$, yielding the highest AUC scores of 99.02% on Ped2 and 75.51% on ShanghaiTech. Increasing σ_c and σ_s to 1.0 leads to a noticeable performance drop, indicating that overly large values weaken the model's ability to effectively separate anomalous from normal patterns. Conversely, very small values result in a slight degradation, likely due to insufficient enforcement of the compactness–separation trade-off. Overall, these results show that ABMA is moderately sensitive to σ_c and σ_s, and the configuration $(0.1, 0.1)$ achieves the optimal balance.

6.5. Past-future weights

The motivation for adopting a bidirectional structure stems from the observation that normal object motions are typically smooth in both temporal directions, while anomalies disrupt this continuity. To validate this, we conducted an ablation study on the past–future weights (σ_p, σ_f) used in Eqs. (17) and 26. Specifically, σ_p controls the past loss contribution, and σ_f adjusts the future loss. We evaluated eleven configurations from pure past (1.0, 0.0) to pure future (0.0, 1.0) in 0.1 increments.

Table 12
Average and maximum AUC scores (%) over 5 runs of ABMA with different values of past–future weights (σ_p, σ_f) on the Ped2 dataset. Boldface indicates the best result.

(σ_p, σ_f)	Ped2 avg/max
(1.0, 0.0)	88.55/93.26
(0.9, 0.1)	93.96/95.65
(0.8, 0.2)	95.17/97.49
(0.7, 0.3)	95.77/97.51
(0.6, 0.4)	96.38/97.71
(0.5, 0.5)	**98.24/99.02**
(0.4, 0.6)	96.41/98.31
(0.3, 0.7)	96.06/97.84
(0.2, 0.8)	95.66/97.41
(0.1, 0.9)	95.48/96.82
(0.0, 1.0)	89.11/91.52

Table 13
Average and maximum AUC scores (%) over 5 runs of ABMA with the different values of memory items N on Ped2 and ShanghaiTech datasets. Boldface indicates the best result.

N	Ped2 avg/max	ShanghaiTech avg/max
5	95.83/98.31	72.71/73.03
10	**98.24/99.02**	**73.78/75.51**
20	97.79/98.14	70.29/71.76
50	95.46/98.23	72.80/74.22
100	97.83/98.64	72.32/72.86

As shown in Table 12, the balanced setting (σ_p, σ_f) = (0.5, 0.5) achieves the best performance, with an average AUC of 98.24% and a maximum of 99.02% on Ped2. Increasing the weight on the past direction improves the performance from 88.55% at (1.0, 0.0) to 96.38% at (0.6, 0.4), indicating that preceding frames provide essential temporal cues. However, relying solely on past information limits temporal context and reduces generalization. Shifting the weighting toward the future causes the AUC to drop from the peak at (0.5, 0.5) to 95.48% at (0.1, 0.9) and 89.11% at (0.0, 1.0), suggesting that future-only prediction becomes unstable without past dynamics as an anchor. Overall, the results show that combining both temporal directions best captures normal temporal consistency and empirically supports the bidirectional memory design in ABMA.

6.6. Number of memory items

The number of memory items N determines the capacity of ABMA to store prototypical normal patterns in both the past and future directions. To evaluate the sensitivity of this hyperparameter, we conducted ablation experiments on Ped2 and ShanghaiTech. As shown in Table 13, setting $N = 10$ achieves the highest average AUC scores over five runs—98.24% on Ped2 and 73.78% on ShanghaiTech—surpassing $N = 5$ by 2.41% and 1.02%, respectively. In contrast, larger settings ($N = 20$, 50, 100) yield lower performance, with average drops of up to 2.78% on Ped2 and 3.49% on ShanghaiTech, suggesting redundancy and overfitting in complex scenes. These findings confirm $N = 10$ as the optimal configuration for the ABM module in ABMA.

6.7. Anomaly score parameter

The parameter α controls the balance between image-level and feature-level errors in computing the final anomaly score. To evaluate

Table 14
Average and maximum AUC scores (%) over 10 runs of ABMA with different values of α in the anomaly score formulation. Values are reported as mean ± standard deviation. Boldface indicates the best result.

α	Ped2	Avenue	ShanghaiTech	IITB-Corridor
0.1	96.57 ± 1.31	86.99 ± 1.41	71.43 ± 1.30	68.51 ± 0.41
0.2	97.22 ± 1.04	87.78 ± 1.20	72.01 ± 0.81	68.92 ± 0.66
0.3	97.58 ± 0.81	88.08 ± 1.03	72.46 ± 0.55	69.17 ± 0.90
0.4	97.80 ± 0.68	**88.16** ± 0.90	72.81 ± 0.49	69.31 ± 1.04
0.5	97.94 ± 0.67	88.10 ± 0.83	73.09 ± 0.57	**69.34** ± 1.07
0.6	**98.00** ± 0.72	87.93 ± 0.79	73.29 ± 0.67	69.27 ± 1.00
0.7	97.99 ± 0.79	87.62 ± 0.75	73.43 ± 0.77	69.10 ± 0.83
0.8	97.90 ± 0.90	87.14 ± 0.68	73.51 ± 0.84	68.85 ± 0.59
0.9	97.58 ± 1.00	86.43 ± 0.50	**73.52** ± 0.89	68.53 ± 0.34

its impact, we varied α from 0.1 to 0.9 in increments of 0.1 and reported the average AUC results over ten runs on four benchmark datasets (Table 14), with values are reported as mean ± standard deviation. The optimal α values for the single-scene datasets Ped2, Avenue, and IITB-Corridor are 0.6 (with AUC = 98.00% ± 0.72%), 0.4 (with AUC = 88.16% ± 0.90%), and 0.5 (with AUC = 69.34% ± 1.07%), respectively, indicating that a balanced combination of appearance- and feature-level cues is generally beneficial. In contrast, ShanghaiTech achieves its highest AUC (73.52% ± 0.89%) at $\alpha = 0.9$, which can be attributed to its multi-scene nature, where image-level reconstruction errors become more informative. Overall, the mean of the best-performing settings across datasets is about 0.6, providing a robust trade-off between low-level appearance and high-level semantic information. Thus, $\alpha = 0.6$ is adopted as the default configuration for ABMA.

7. Visualization analysis

This section visualizes and compares the anomaly scores of ABMA and MNAD [11]. After that, we provide frame prediction examples with error maps and visualize the ability of ABMA in detecting subtle anomalies.

7.1. Anomaly scores

This section visualizes the anomaly scores of ABMA and MNAD [11] on one video from each dataset. Fig. 3 shows the results on the long-duration anomaly datasets. ABMA more effectively distinguishes normal from anomalous frames. During normal segments, it produces consistently lower scores than MNAD, indicating better suppression of false positives and more stable reconstruction of normal patterns. During anomalous events, ABMA yields higher scores, providing a clearer separation between normal and abnormal frames. Another notable observation is the smoother anomaly score curves produced by ABMA compared to those of MNAD. ABMA's scores fluctuate less between consecutive frames, whereas MNAD exhibits greater variability, including sudden spikes and drops, even within regions containing only normal frames. These abrupt changes in MNAD's scores are likely caused by background noise or subtle environmental variations.

Fig. 4 presents the anomaly scores of ABMA and MNAD on the short-duration anomaly datasets. It can be observed that ABMA consistently detects anomalous frames earlier and produces higher anomaly scores than MNAD during abnormal periods. Moreover, ABMA maintains relatively stable scores on normal frames, further highlighting its robustness. Overall, the results in this subsection demonstrate that ABMA generates more reliable and discriminative anomaly scores than MNAD on both long- and short-duration anomaly datasets. This improvement can likely be attributed to the effectiveness of ABMA's new component, i.e., the attention-guided bidirectional memory module.

7.2. Predicted frames and error maps

This subsection further analyzes ABMA's behavior through visualizations of predicted frames and error maps. Fig. 5 shows results from the Ped2, Avenue, ShanghaiTech, and IITB-Corridor datasets. For each dataset, the top row represents past-frame reconstructions, and the bottom row shows future-frame predictions. Within each group, the first and second columns display the ground truth and predicted frames, the third column shows the pixel-wise mean squared error map, and the

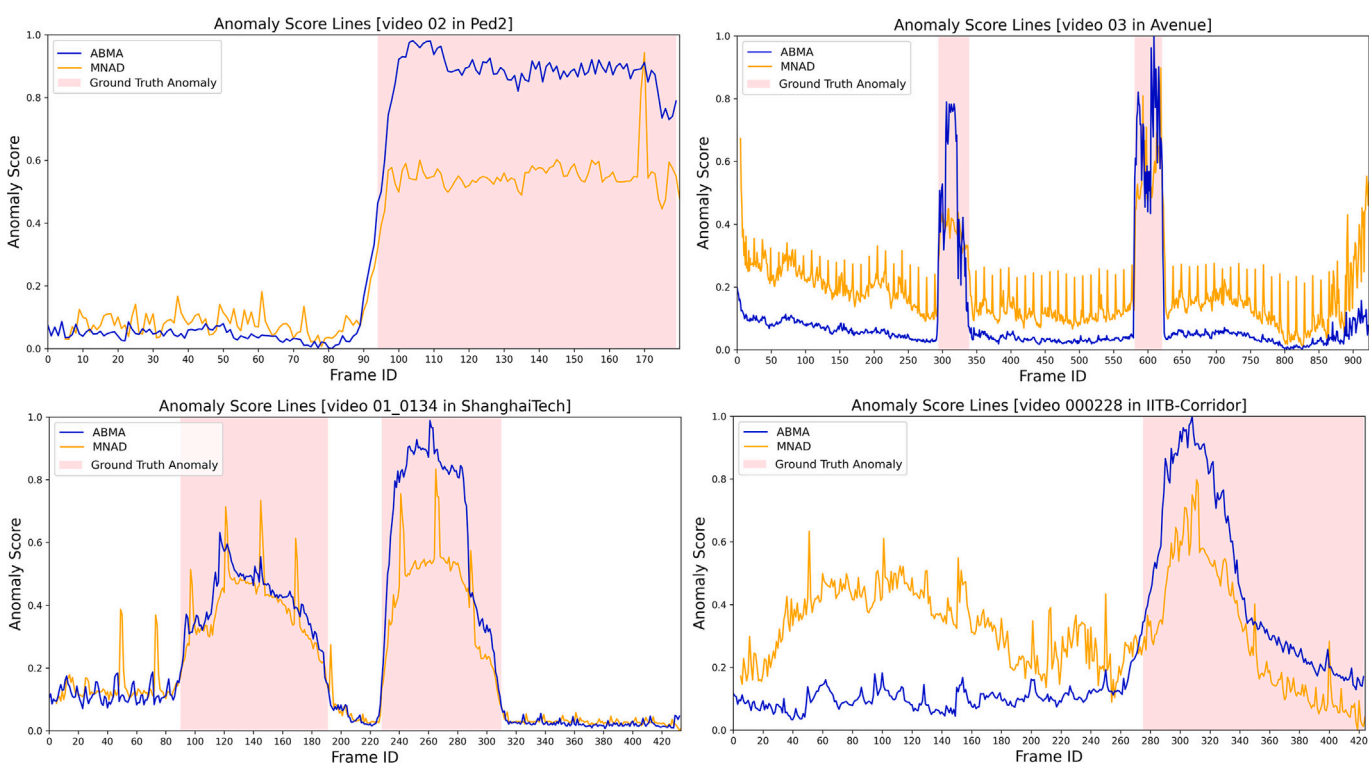

Fig. 3. Anomaly scores of ABMA and MNAD [11] on videos 02, 03, 01_0134, and 000228 from Ped2, Avenue, ShanghaiTech, and IITB-Corridor datasets, respectively.

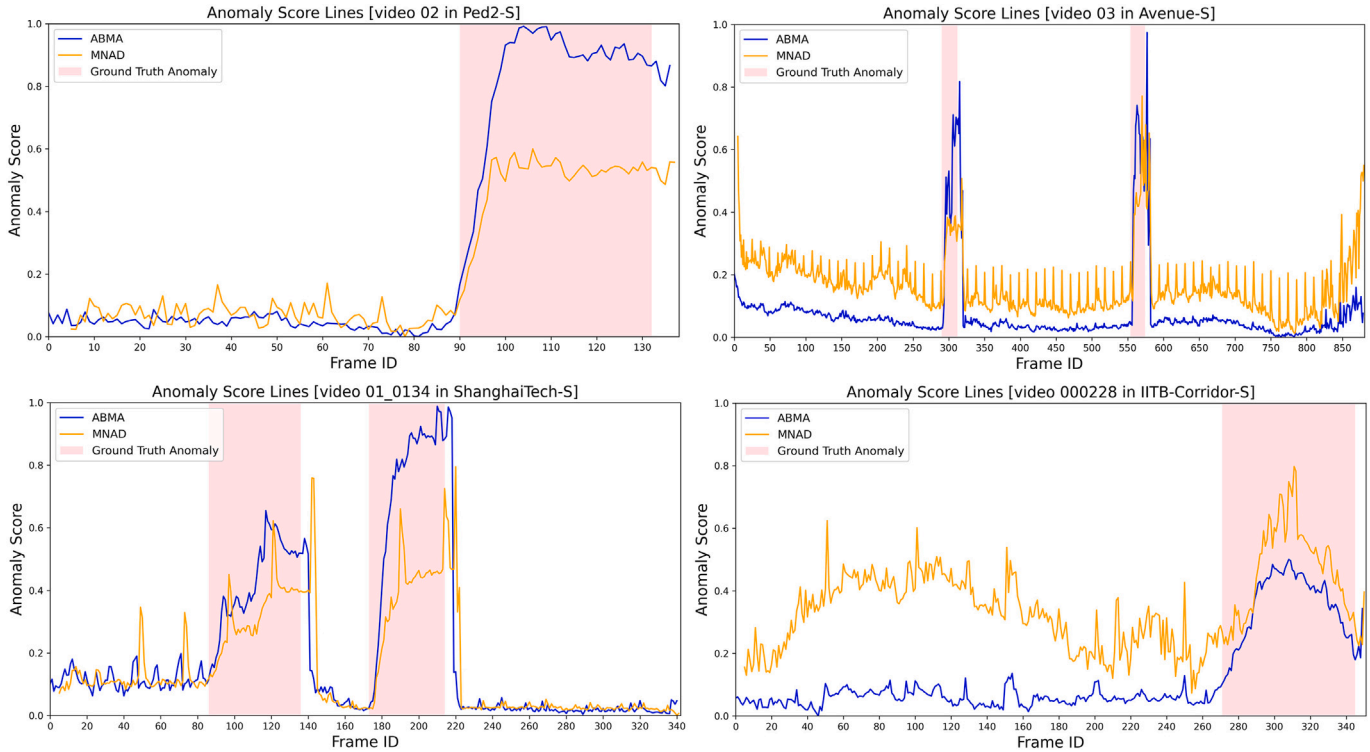

Fig. 4. Anomaly score of ABMA and MNAD [11] on videos 02, 03, 01_0134, and 000228 from Ped2-S, Avenue, ShanghaiTech-S, and IITB-Corridor-S datasets, respectively.

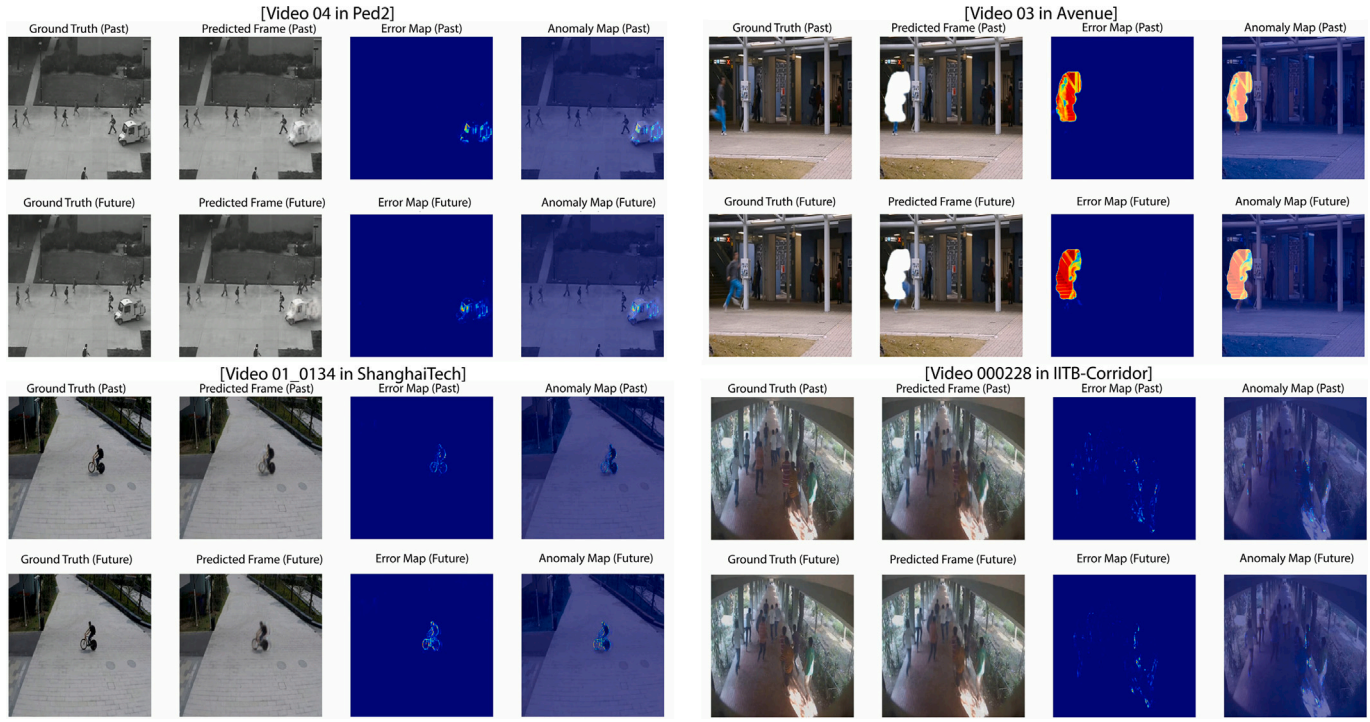

Fig. 5. Visualization of frames from videos 04 (Ped2), 03 (Avenue), 01_0134 (ShanghaiTech), and 000228 (IITB-Corridor). In each group, the first and second row displays the past and the future frames, respectively. Each row include the ground truth frame, the predicted frame, the error map, and the anomaly map, from left to right.

fourth overlays this map onto the ground truth to form the anomaly map.

ABMA clearly highlights abnormal events across all tested videos. In Ped2, it identifies a cart as abnormal, producing large prediction errors. In Avenue, a running person is detected as anomalous through notable frame deviations. In ShanghaiTech, the model captures the irregular motion of a cyclist, evident in localized error patterns. In IITB-Corridor, simultaneous group movement triggers elevated reconstruction errors.

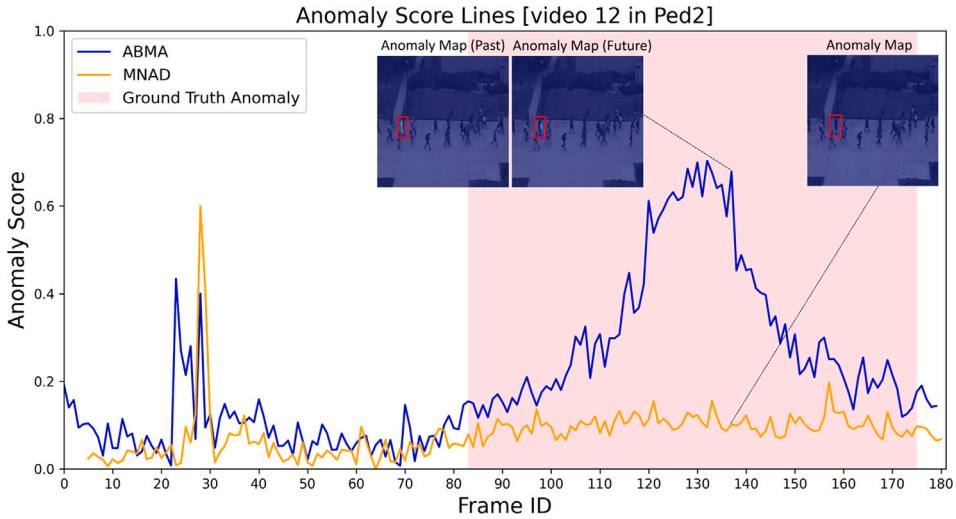

Fig. 6. Anomaly scores of ABMA and MNAD [11] in video 12 on Ped2. The anomaly object is denoted by a red rectangle.

Overall, ABMA effectively reconstructs normal scenes with minimal error but shows strong deviations aligned with abnormal entities or actions, confirming its capability to distinguish normal from anomalous patterns across varied environments.

7.3. Subtle anomalies

This subsection analyzes the ability of ABMA to detect subtle deviations caused by anomalous events. We illustrate this using video 12 from the Ped2 test set, where an anomaly occurs as a skateboarder moves from left to right through a crowd of pedestrians. The challenge in this scene lies in the subtle nature of the deviation—the skateboarder's motion closely matches the walking speed of surrounding pedestrians and is partially occluded by them—making the anomaly difficult to identify. As shown in Fig. 6, MNAD's anomaly scores remain low when the skateboarder appears, indicating its failure to capture subtle deviations. In contrast, ABMA exhibits a sharp score increase as the skateboarder enters, clearly distinguishing the anomalous motion from normal behavior. This demonstrates ABMA's superior ability to detect subtle anomalies often missed by single-memory, unidirectional models like MNAD.

8. Conclusions, limitations and future work

This section summarizes the novelties of the proposed model, discusses its limitations, and outlines potential directions for future improvement.

8.1. Novelties

In this paper, we proposed ABMA, a novel model for video anomaly detection that leverages an attention-based bidirectional memory autoencoder. This design enables ABMA to capture both past and future contexts, thereby enhancing anomaly prediction. Experimental results on four benchmark datasets demonstrate that ABMA consistently outperforms all other evaluated methods. In particular, ABMA shows strong effectiveness in detecting subtle anomalies, such as short-duration anomaly events.

8.2. Limitations

Although ABMA achieves state-of-the-art performance on multiple benchmarks, several failure cases remain. On Avenue, the model struggles with visually entangled anomalies—such as a pedestrian pushing a bicycle in video 16—where normal and abnormal objects overlap spatially. In videos 17 and 18, anomalies with partially visible actions (e.g., hand waving or stamping near the frame edge) provide weak temporal cues. On ShanghaiTech (e.g., video 10_0074), failures occur when anomalies appear distant from the camera or remain stationary, making them visually similar to normal behavior. These results highlight challenges for ABMA, including occlusion, slow or static anomalies, and large scale variations.

8.3. Future directions

Several promising directions for future research emerge from this work. First, we plan to examine the effectiveness of ABMA when incorporating more diverse types of pseudo-anomalies. Second, we will explore alternative memory strategies, such as dynamic memory [51] and hierarchical memory [10], to further enhance temporal modeling. Last but not least, we aim to explore dynamic spatial-temporal attention and multi-view fusion strategies to address the failure cases encountered by ABMA.

CRediT authorship contribution statement

Anh Le: Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Quang Uy Nguyen:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Quang Uy Nguyen reports that financial support was provided by the Vingroup Innovation Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets used in this study are publicly available and commonly used in the video anomaly detection community. Detailed information can be found in the original papers describing these datasets.

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